



AIT401-QP2 MAY 2024 - PREVIOUS YEAR QUESTION PAPER

Foundations of Deep Learning (APJ Abdul Kalam Technological University)



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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree (S, FE) Examination May 2024 (2019 Scheme)

Course Code: AIT401**Course Name: FOUNDATIONS OF DEEP LEARNING****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- | | | |
|----|---|-----|
| 1 | Discuss the importance of validation sets in training neural networks. | (3) |
| 2 | Describe how the softmax activation function works? | (3) |
| 3 | Conceptualize Ada-grad scheme for optimization. | (3) |
| 4 | Suppose you have a 3-dimensional input $x = (x_1, x_2, x_3) = (4, 4, 2)$ fully connected with weights (0.5, 0.3, 0.2) to one neuron which is in the hidden layer with sigmoid activation function. Calculate the output of the hidden layer neuron. | (3) |
| 5 | Why do we use pooling in convolutional neural networks? Illustrate with an example how pooling works? | (3) |
| 6 | Define the concept of the receptive field. Mention two strategies for expanding the receptive field without increasing the filter size. | (3) |
| 7 | Draw the circuit diagram and unfolded computational graph of a simple RNN with no outputs. | (3) |
| 8 | Differentiate between LSTM and GRU. | (3) |
| 9 | Define under-complete autoencoder. Explain its significance in feature extraction and the role of the loss function in training. | (3) |
| 10 | List the difference between Boltzmann Machine and Deep Belief Network. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | a) Discuss the practical issues in neural network training | (10) |
| | b) Describe the steps involved in training a neural network. | (4) |

OR

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|----|---|-----|
| 12 | a) What is backpropagation, and how is it used in training neural networks? | (7) |
| | b) What are activation functions? Explain different activation functions (Identity, Sigmoid, Tanh, and ReLU) and demonstrate the ability to select appropriate ones based on specific problem scenarios within neural network design. | (7) |

Module II

- 13 a) How does the vanishing gradient problem impact the learning process in a neural network? Illustrate this issue using a basic network diagram. (7)
b) Explain parameter specific optimization algorithms (Adam and RMSProp) (7)

OR

- 14 a) Discuss the significance of dropout layers in network architecture. (4)
b) Draw the architecture of AlexNet and explain its distinctive features. (10)

Module III

- 15 a) Explain the architecture of a Convolutional Neural Network (CNN) and its fundamental components. (8)
b) Discuss different formats of data that can be used with CNN. (6)

OR

- 16 a) Provide examples of diverse applications where Convolutional Neural Networks excel and explain their effectiveness in those domains. (10)
b) Consider an input image with dimensions of 64 x 64 pixels. You apply a convolutional operation with a kernel (filter) size of 5x5, a padding of 2, and a stride of 1. Calculate the dimensions of the output feature map. Additionally, determine the padding value if we need the output to have the same size as the input with a stride of 2. (4)

Module IV

- 17 a) Describe an encoder - decoder sequence to sequence architecture which takes a sequence $[x_1, \dots, x_n]$ as input and generates a sequence $[y_1, \dots, y_m]$ as output for machine translation? (7)
b) Explain Gated Recurrent Units. (7)

OR

- 18 a) Explain the concept of computational graphs in the context of Recurrent Neural Networks (RNNs). How are they relevant in understanding RNN computations? (8)
b) Discuss the challenges of training RNN. (6)

Module V

- 19 a) Explain the concept of Autoencoders and their role in unsupervised learning. (7)
b) Explain the concept of adversarial training and the interplay between the generator and discriminator. How do GANs generate realistic synthetic data? (7)

OR

- 20 a) Explain the architecture and functioning of Deep Belief Networks (DBNs). (7)
b) Discuss the applications of Autoencoders. (7)
